

EXPERT INSIGHT

Time Series with PyTorch

Modern Deep Learning Toolkit for Real-World Forecasting Challenges



Graeme Davidson
Lei Ma

<packt>

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To Claire and Sebastian, every day with you both is a blessing that lightens every struggle and gives it purpose.

– Graeme Davidson

To Han and Yoyo, for this beautiful life together.

– Lei Ma

Acknowledgments

During my (Graeme Davidson) undergraduate degree in the late 2000s, I encountered a BBC documentary which discussed demand forecasting within Tesco. The scale at which they

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were working was incredible to me; it still is. Unfortunately, the documentary only superficially covered how statistics were applied to prediction, but I was fascinated by how we apply statistics over time. Why don't many process, or research experiments account for the context and passage of time? Clearly, the answer was that time series forecasting is difficult. Of course, data management plays a huge role in that. Around the same period, every day, the news was full of time series as financial markets crashed globally, in part because people who paid a lot of money to analyze market data were not paying attention to the details of their data, along with several other factors. But overall, this is what Michael Burry capitalized on in 2007.

Several years have passed since then, and I would say that time series forecasting is somewhat easier to implement than it was. It does help that one can now spin up an autoregressive foundation model to churn out code, but as to how well they do this, I think they still leave a lot to be desired. And it is increasingly obvious that *vibe coding* undermines learning and understanding. That isn't to say you shouldn't use them, you should! But rather that you should make sure you understand what you are doing. Beforehand, I used to search the internet trawling for people who knew what they were talking about: Abhishek Thakur, Rob Hyndman, Stephan Kolassa, and Konrad Banachewicz, etc. In particular, around the time I started in demand forecasting, Abhishek Thakur hosted a series of YouTube videos on time series forecasting by Konrad Banachewicz. A few years later, Packt contacted me on LinkedIn to ask if I'd write a book on time series forecasting. I certainly wasn't as skilled then, so I suggested they contact Konrad. For some reason they then came back to me and said that Konrad would like to work on the book but needed a co-author. I wasn't going to turn down the opportunity to work with a Kaggle grandmaster. Unfortunately, as time went on, Konrad became too busy with other projects and work to continue. However, during my online searches, I came across Lei Ma's website; his approach to deep learning and modeling in general appealed to me. He was precise and concise, providing intuition about theory that so many blogs fail to convey; of course, he is a physicist by trade. Luckily, I was able to persuade him to take on the project.

Lei and I have very different writing styles, in part due to our backgrounds (I have a cognitive neuroscience background). While he provides the intuition of a physicist and tends to draw out how deep learning systems make sense of data, I tend to focus on systems and experimentation. I believe that this gives you the best of both perspectives so that you can confidently design and test your own forecasting systems, with the understanding you need to make good decisions.

I hope you enjoy the book!

Graeme Davidson

Contributors

About the authors

Graeme Davidson is a Lead Data Scientist at Retail Express, where he redesigned the company's demand forecasting framework in line with contemporary statistical learning practices. His background spans cognitive neuroscience, researching implicit reward processing and human decision-making, through advertising analytics to research-focused demand forecasting. He is an active contributor to several data science Slack and Discord communities, an occasional competitor in forecasting competitions, and was approached by Packt in late 2022 to write the book he wished had existed when he first fell down an ARIMA rabbit hole chasing answers about how supermarkets actually forecast demand, and how a quantitative researcher models financial markets.

I would like to thank the team at Packt who worked on this book: Ayushi Bulani, Shashank Desai, and Seemanjay Ameriya. A special thank you to Lei Ma for taking on this project as co-author, and to Joe Xiao, whose technical review genuinely strengthened our work. Thank you to my brother Andrew and my best friend Ben Goodhart for their encouragement throughout, and to my 'sweat the death out' group — you know who you are. I owe a great deal to the managers I have had over the years: Gareth Jenkins for his patience, Andy Mahood for changing my perspective, and Simon Duckworth, who was my manager when this project began in 2023 and actively promoted my development throughout. I would also like to thank Tim Kirkham for developing my critical reasoning skills during my PhD — I am still, and always will be, grateful. Finally, thank you to my family for accepting that this book consumes our valuable time together. Claire and Sebastian, none of this happens without you.

Lei Ma is a physicist-turned data scientist specializing in time series forecasting. He is a theorist but has tackled real-world forecasting challenges across a variety of industries like housing, logistics, ecommerce, and manufacturing. Lei has led and delivered numerous forecasting projects where he combines deep expertise in building advanced time series models with a strategic approach to delivering holistic business insights. Lei creates time series forecasting tutorials online and joined the venture when Graeme approached him to collaborate on this book.

I would like to thank Graeme Davidson for this exciting and fun collaboration, as well as the Packt team — Ayushi Bulani, Shashank Desai, Seemanjay Ameriya, Tushar Gupta, and everyone who worked on this book — for all their guidance and support. The reviewers, Jack Lee and Joe Xiao, have shaped the chapters with their expertise, and I would like to express my sincere thanks to them. I also want to express my gratitude to the open-source community for driving tech forward and providing the foundational knowledge for this book. Finally, thank you to my wife, Han Lu, for being the driving

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force that pushed me past every doubt, and to our daughter, Yoyo, for the peaceful nights and joyful days that made this journey enjoyable.

About the reviewers

YiZi (Joe) Xiao, Ph.D., is a Senior Principal AI/ML Scientist at Optum (UnitedHealth Group), where he leads teams developing machine learning and LLM-driven solutions in healthcare. He has over a decade of experience in AI and Data Science across industry and academia, including roles at Target, General Mills, Starkey Hearing Technologies, and Medtronic. Dr. Xiao holds a Ph.D. in Biomedical Engineering from the University of Minnesota, an M.S. in Computer Science from the University of Illinois Urbana-Champaign, and a B.S. in Bioengineering from UC Berkeley. He has authored numerous publications and patents and serves as a reviewer for leading journals in biomedical engineering.

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